LESSON PLAN

NDT (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Magnetic particle inspection application advantages, limitations, circulars and longitudinal magnetization, general procedure for magnetic particle inspection
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Magnetic particle inspection application advantages, limitations, circulars and longitudinal magnetization, general procedure for magnetic particle inspection various types of parts which can be inspected after inspection reasons for demagnetization
5. Specific Objectives	19.1.1.1 Trainee will be able to describe types of NDT and perform liquid penetrant, magnetic particles and eddy current inspections

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of longitudinal and circular magnetization using yoke and coil; safety precautions regarding electric shock and eye protection; instructor supervision during application on test blocks.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) List three common types of NDT methods.
(b) Why is magnetic particle inspection preferred for ferromagnetic materials?
(c) What is the primary difference between liquid penetrant and magnetic particle inspection?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Application advantages and limitations of MPI • Circular and longitudinal magnetization principles • General procedure for magnetic particle inspection	L & Ds	PPT, IB	15 min
(b)	• Types of parts suitable for MPI inspection • Post-inspection reasons for demagnetization • Interpretation of indications and false calls	L & Ds	PPT, PDF	20 min
(c)	• Demonstration of yoke magnetization technique • Demonstration of coil magnetization technique • Safety protocols for magnetic particle testing	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between circular and longitudinal magnetization.
(b) List two reasons why a part might require demagnetization after inspection.
(c) Describe the general steps involved in the magnetic particle inspection procedure.

12. Home Assignment and Follow Up	Prepare a brief summary comparing the advantages and limitations of Magnetic Particle Inspection with the upcoming Eddy Current Inspection topic in LP 04.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Eddy current inspection, principle of operation, function of basic system, coil impedance, electrical conductivity mag permeability, types of inspection coils, application of various types of parts	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min